PROJECT OVERVIEW

Research and Understanding Quantum Computing

- **Explored the fundamentals of quantum computing, including qubits, superposition, and probability.**
- **Investigated Bob Coecke's concept of Quantum Picturalism and the ZX-calculus as alternative ways of representing quantum logic.**
- **Used Python and Qiskit to simulate basic quantum gates and circuits, building an understanding of how quantum operations work.**

Development of the Quantum Logic Visualiser

- **Designed and programmed a Python application that converts quantum circuits into visual representations.**
- **Created a custom system of coloured "spiders" based on ZX-calculus structures to represent quantum operations and connections between qubits.**
- **Built a backend that translates quantum gates into connected visual nodes and developed a Tkinter interface to display these diagrams in a simple, beginner-friendly way.**

Results and Educational Impact

- **Demonstrated that quantum circuits can be represented pictorially rather than solely through mathematical equations.**
- **Showed how interactive visualisations help users understand what quantum gates do and how they affect qubit states.**
- **Concluded that visual reasoning can make key quantum computing concepts more accessible to students and beginners, supporting learning and engagement.**